cuts data transfer costs and keeps private data on the device. Compared with commercial edge platforms, it is free, vendor-neutral and governed openly under LF Edge, so there is no lock-in.

Key components of Open Horizon

Component	Where it runs	What it does
Exchange	Management hub	Stores users, organisations, nodes, services, policies and patterns; exposed as a REST API on port 3090
Agreement bot (agbot)	Management hub	Finds nodes that match a deployment policy and proposes agreements to them
Cloud Sync Service (CSS)	Management hub	Distributes files and machine learning models to edge nodes (port 9443)
Agent (anax)	Each edge device or cluster	Accepts agreements, pulls containers, runs and monitors them
hzn CLI	Any admin or developer machine	Command-line tool for publishing, registering and monitoring

Open Horizon combines a central control plane with autonomous edge agents. Its key features are:

Policy-based deployment: Nodes advertise properties such as *hasCamera=true* or *location=factory-3*. Deployment policies state constraints, and the agbot matches them automatically, so a new device that meets the rules receives its workload without any manual step.

Patterns: For simple fleets, a pattern names a fixed list of services that a node runs when registered with it.

Autonomous agents: Each agent negotiates, deploys and heals its own workloads and keeps running when disconnected from the hub.

Model management system: Machine learning models and configuration files are updated separately from the container that uses them, so you can retrain a model without rebuilding the service.

Edge devices and edge clusters: The same hub manages Docker-based devices and Kubernetes clusters such as K3s, MicroK8s and Red Hat OpenShift.

Security: Service definitions can be cryptographically signed, agreements are encrypted, multi-tenancy isolates organisations, and a secrets manager delivers credentials to services at run time.

Rollback and agent upgrades: Services can fall back to a previous version if an update fails, and node management policies upgrade the agents themselves.

Zero-touch onboarding: FIDO Device Onboard (FDO) support lets new devices register when first powered on.

Prerequisites for installing Open Horizon

The simplest setup is the all-in-one install, which puts the management hub, an agent and the CLI on one machine. You need the following before you start:

Item	Requirement
Operating system	Ubuntu 24 on an x86_64 machine for the all-in-one hub; edge agents also run on ARM boards such as Raspberry Pi
Access	A root account or sudo rights
Memory and disk	At least 4GB RAM and 20GB free disk is a practical minimum for the hub (suggested, not an official figure)
Container engine	Docker (installed by the script if missing); Podman works if you set *DOCKER_ENGINE=podman*
Tools	*curl*, *git*, *jq* and *make*
Network	Internet access to download images, and a fixed IP address that edge devices can reach
Open ports on the hub	3090 (Exchange), 3111 (agbot), 9443 (CSS)
Accounts	A Docker Hub (or other registry) account to push your own service images

Decide on the hub's network-facing IP address before installing. The script binds the hub to *HZN_LISTEN_IP;* if you skip this, only the hub machine can connect.

Installing Java and setting up the environment

Open Horizon itself does not need Java on the host. The agent and the *hzn* CLI are written in Go, and the Exchange, which runs on the Java Virtual Machine, ships inside its own container. Install a Java Development Kit (JDK) only if you plan to write your edge services in Java or Spring Boot.

Step 1: Update the system and install base tools

```
sudo apt update && sudo apt -y upgrade
sudo apt install -y curl git jq make
```

Step 2: Install Java (optional, for Java-based services)

```
sudo apt install -y openjdk-21-jdk
java -version
```